Checkpoint 3: An Order-Controlled Search Agent for Misstatement-Risk Screening

Ryan Khan IDAI-780, Fall 2026 Instructor: Zhiqiang Tao Code: github.com/x4teen/fraud-cot-search-agent

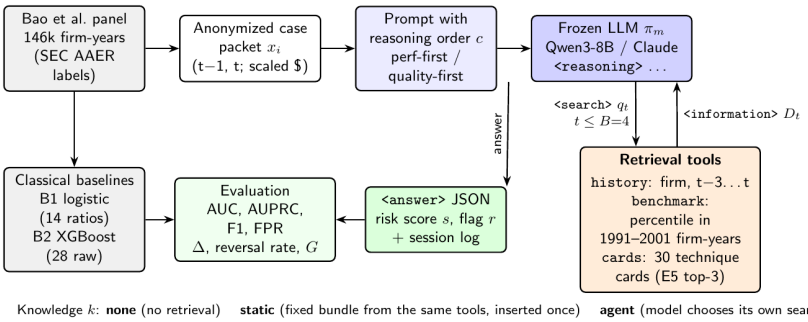


Figure 1: Framework. A frozen LLM reasons in order c and, in the agent setting, issues up to $B{=}4$ searches to three leakage-safe tools before answering; outputs are scored with the same metrics as the classical baselines.

1 Formulation

Data and task. We use the Bao et al. [1] panel of U.S. firm-years with SEC AAER labels $y_i \in \{0, 1\}$. Splits are temporal: train 1991–2001, test 2003–2008, with a two-year gap and serial-fraud relabelling. Case i is an anonymized packet x_i with 28 items for years $t{-}1, t$ and 14 ratios. Firm ID and calendar years are removed, and dollar amounts are multiplied by a random constant $\lambda_i \in [0.5, 2]$.

Inference model. A frozen LLM π_m ($m \in$ {Qwen3-8B, Claude Sonnet 5.5}) runs under reasoning order $c \in \{\mathrm{D}, \mathrm{P}, \mathrm{Q}\}$: direct; *performance-first* (growth, margins, ROA, the headline story); and *quality-first* (accruals, cash, receivables, inventory, soft assets). The two CoT prompts contain the same words in a different order. Knowledge is $k \in \{\varnothing, \mathrm{S}, \mathrm{A}\}$: none, static bundle, or search agent. Following Search-R1 [2], without RL, the agent produces

$$\pi_m(\tau \mid x_i; c) = \prod_{t=1}^{T} \pi_m(t_t, q_t \mid x_i, c, \tau_{<t}), \quad T \le B, \tag{1}$$

where t_t is a `<reasoning>` step and q_t a `<search>` call to one of three tools. `history` returns the firm's own $t{-}3..t$ values scaled by λ_i. `benchmark` returns the percentile of a ratio among 1991–2001 firm-years. `cards` returns the top-3 of 30 technique cards,

$$D_t = \mathrm{TopK}_{d \in \mathcal{T}} \cos\big(\phi(q_t), \phi(d)\big), \tag{2}$$

where ϕ is the E5 encoder. Each D_t is inserted inside `<information>`; the model conditions on it but does not generate it. The answer gives a risk score $s_i^{m,c,k} \in [0, 100]$ and a red flag $r_i = \mathbb{1}[s_i \ge \theta_{m,c,k}]$, with θ chosen to maximize F1 on a disjoint dev split (2000–01 cases).

Hypotheses. *H1 (order bias):* $\Delta_i^{m,k} = s_i^{m,\mathrm{P},k} - s_i^{m,\mathrm{Q},k}$ is negative, i.e. reasoning that starts from performance anchors on the headline numbers. We report mean Δ, $|\Delta|$, the red-flag reversal rate ρ, and a Wilcoxon signed-rank test. *H3 (knowledge):* $G^{m,c} = \mathrm{AUPRC(A)} - \mathrm{AUPRC(S)} > 0$, and $|\Delta^{\mathrm{A}}| < |\Delta^{\varnothing}|$.

Algorithm 1: Order-controlled search agent (batched)

Input: cases $\{x_i\}$, model π_m, order c, budget $B{=}4$
$h_i \leftarrow \textsc{Prompt}(x_i, c)$ for all i; active $\leftarrow$ all;
while *active* $\neq \varnothing$ *and step* $< B{+}4$ **do**
  $y_i \leftarrow \pi_m(h_i)$ until `</search>` | `</answer>`; `// batched`
  foreach *active* i **do**
    if `<answer>` *in* y_i **then**
      close i
    else if $(tool, q) \leftarrow \textsc{Parse}(y_i)$, $n_i < B$ **then**
      $D \leftarrow \textsc{Tool}(tool, q)$;
      $h_i \mathrel{+}= y_i \| $`<information>`$D$`</information>`; log $(q, \mathrm{IDs}(D))$; $n_i{+}{+}$
    else "rethink" once (or budget used): force `<answer>`;
  end
end
return $\textsc{ParseJSON}(h_i)$ for all i;

2 Comparison to Baselines

Model	Condition (c, k)	AUC	AUPRC	$\lvert\Delta\rvert$
B0	random	0.58	0.57	–
B1	logistic, 14 ratios	0.62	0.59	–
B2	XGBoost, 28 items	**0.76**	**0.76**	–
Qwen3-8B	D, ∅ (CP2 Direct)	0.53	0.53	–
	P / Q, ∅ (CoT)	0.57 / 0.61	0.55 / 0.58	3.8
	P / Q, S (static RAG)	0.51 / 0.45	0.52 / 0.50	6.6
	P / Q, A (agent)	0.51 / 0.53	0.51 / 0.52	5.6
Claude Sonnet 5.5	D, ∅	0.53	0.57	–
	P / Q, ∅ (CoT)	0.53 / 0.53	0.54 / 0.57	4.2
	P / Q, S (static RAG)	0.54 / 0.54	0.59 / 0.54	4.1
	P / Q, A (agent)	0.56 / 0.55	0.58 / 0.60	5.4

Table 1: The same 60 balanced out-of-sample test cases (30 later-AAER). Values have 95% case-bootstrap CIs (in repo). On the full test set (35,166 firm-years) B2 reaches AUC 0.725 and AUPRC 0.015 against a prevalence of 0.0074, in line with Bao et al.'s RUSBoost (≈0.71).

Findings. (i) B2 remains best. Every LLM setting has an AUPRC CI that includes the random level (0.50); the best are Qwen quality-first CoT (AUC 0.61 vs. 0.53 Direct) and Claude quality-first agent (AUPRC 0.60 [0.43, 0.75]). (ii) *H1:* the order effect appears only when the model searches. Mean Δ with the agent is -2.6 for Qwen ($[-5.7, 0.0]$, $p{=}0.29$) and -2.0 for Claude ($[-4.0, -0.1]$, Wilcoxon $p{=}0.04$, uncorrected for six tests); without retrieval it is near zero for both (-0.1, $+0.5$). (iii) *H3 is not supported:* no agent-vs-static gain G has a CI that excludes zero (Claude quality-first $+0.05$ $[-0.03, 0.12]$); for Qwen, retrieval raises $|\Delta|$ ($3.8 \to 5.6$–6.6). (iv) *Mechanism:* the order steers the searches. Quality-first raises the share of lookups on earnings-quality items (Qwen 40% → 46%, Claude 43% → 56%), and Claude searches 70% more (1.05 vs. 0.62 per case). Limits: $n{=}60$; Qwen uses 3–5 distinct scores.

3 Implementation Details

Training (B2). XGBoost minimizes $\sum_i w_{y_i} \ell(y_i, f(z_i)) + \Omega(f)$ with $w_1 = n_0/n_1$. Hyperparameters are chosen on fit 1991–97 → validation 2000–01 over depth $\{3, 4, 6\}\times$ trees $\{200, 400\}$ (learning rate 0.05, row and column subsample 0.8). The selected setting (depth 3, 200 trees) is refit on 1991–2001 with median imputation and 1/99% winsorizing fit on training rows only. Per-round loss/AUC/AUPRC and checkpoints every 50 rounds are saved; test years are scored once. **LLM testing.** Qwen3-8B runs in bf16 with Transformers on one A100 (80 GB), with greedy decoding, built-in thinking off, at most 1,024 new tokens per step, and batched lock-step rollouts (16 sessions per batch). Claude runs the same tools through native API tool use at effort low (D) or high (CoT), since it does not write out its reasoning. Each session logs prompt hash, queries, retrieved IDs and raw output; 16 unit tests cover parsing, budget, leakage, determinism and prompt symmetry. Uncertainty comes from 1,000 case-bootstrap resamples, paired across conditions.

References

[1] Y. Bao et al. Detecting accounting fraud in publicly traded U.S. firms using a machine learning approach. *J. Accounting Research*, 58(1), 2020.

[2] B. Jin et al. Search-R1: Training LLMs to reason and leverage search engines with reinforcement learning. *COLM*, 2025.